

CO2xN: Porewater SO4/Cl

May 2026 Samples

2026-07-14

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##Add Required Packages

0.1 Run Information

```
##### Run information - PLEASE CHANGE
Date_Run = "2026-07-07" #Date that instrument was run
Run_by = "Victoria Ellis" #Instrument user
Script_run_by = "Victoria Ellis" #Code user
sample_year <- "2026"
sample_month <- "May"
Site = "GCRew"
Project_name = "CO2xN"
run_notes = "One bad DUP from C1, two bad DUPS from S04.
One bad SPK. Standard curve looks mostly good but the first standards are still low (despite remaking t
We might need to re-purchase the standards.
" #any notes from the run

##### File Names - PLEASE CHANGE
#file path and name for raw summary data file
raw_file_name_c1 = "Raw Data/CO2xN_202605_C1.txt"
raw_file_name_so4 = "Raw Data/CO2xN_202605_S04.txt"

#file path and name of processed data file
processed_file_name = "Processed Data/GCRew_CO2xN_PW_Processed_C1_S04_202605.csv"

##### Log Files - PLEASE CHECK
#downloaded metadata csv - downloaded from Google drive as csv for this year
Raw_Metadata = here("../", "GCRew", "GCRew_Project_Treatment_Metadata.csv")

#qaqc log file path for this year
Log_path = "Raw Data/COMPASS_Synoptic_C1_S04_QAQClog_2026.csv"
```

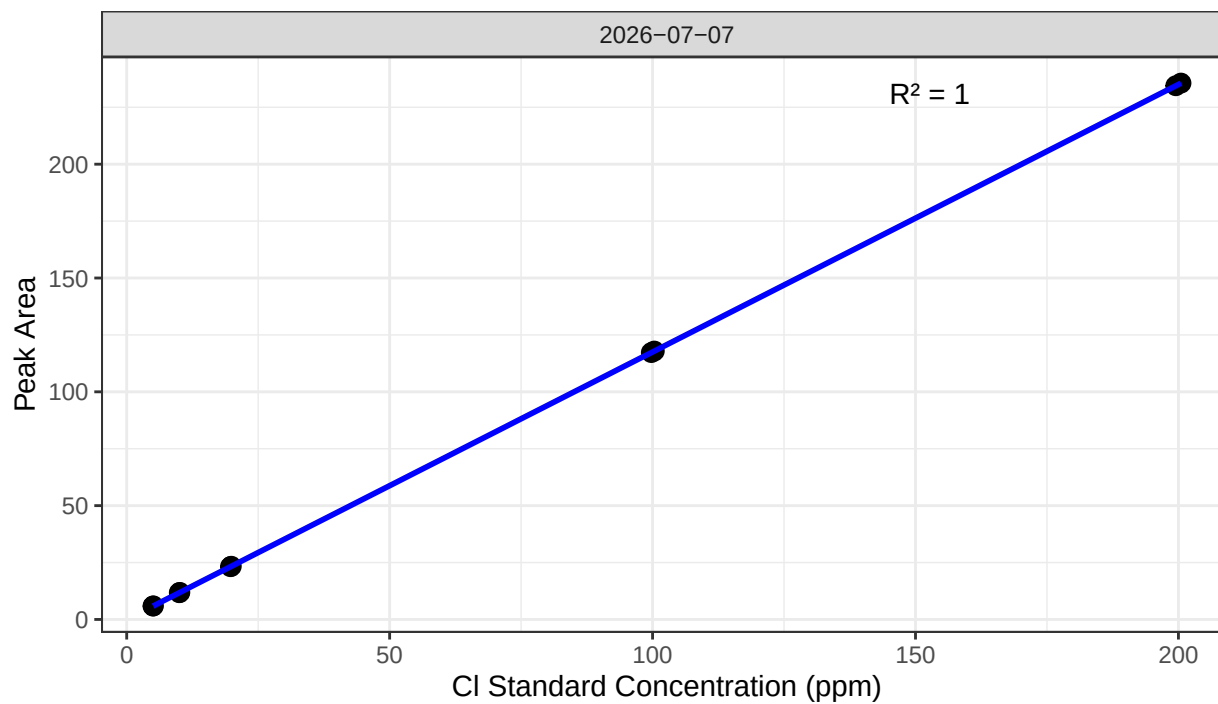
##Set Up Code - constants and QAQC cutoffs

##Read in metadata and create similar sample IDs for matching to samples

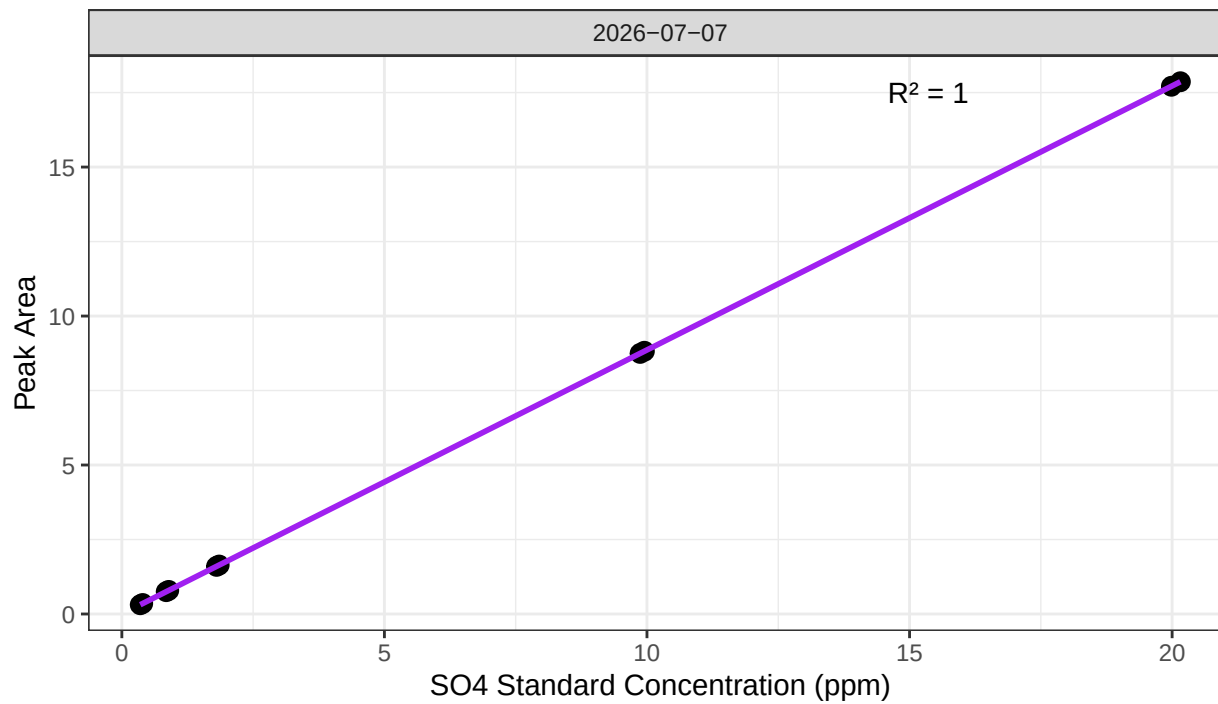
##Import Sample Data

0.2 Assess Standard Curves

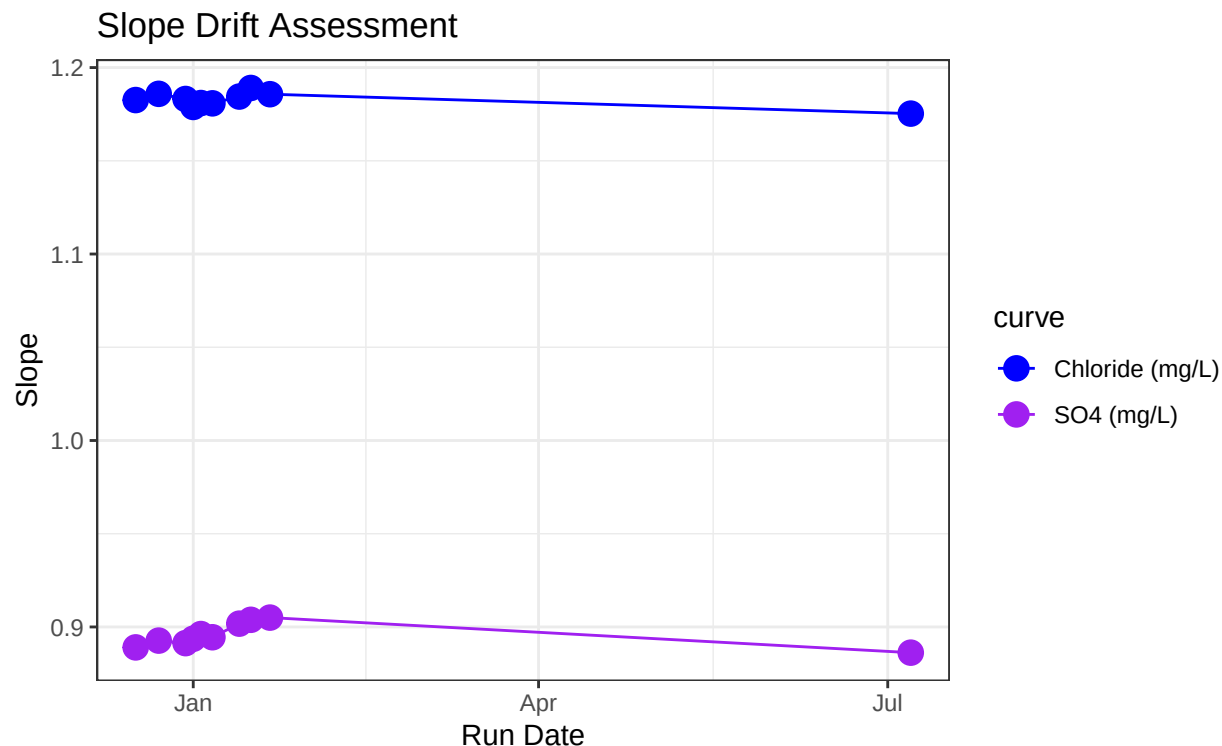
Chloride Std Curve



Sulfate Std Curve



```
## [1] "QAQC log file exists and has been read into the code."
```



```
## [1] "Cl Curve r2 GOOD"
```

```
## [1] "SO4 Curve r2 GOOD"
```

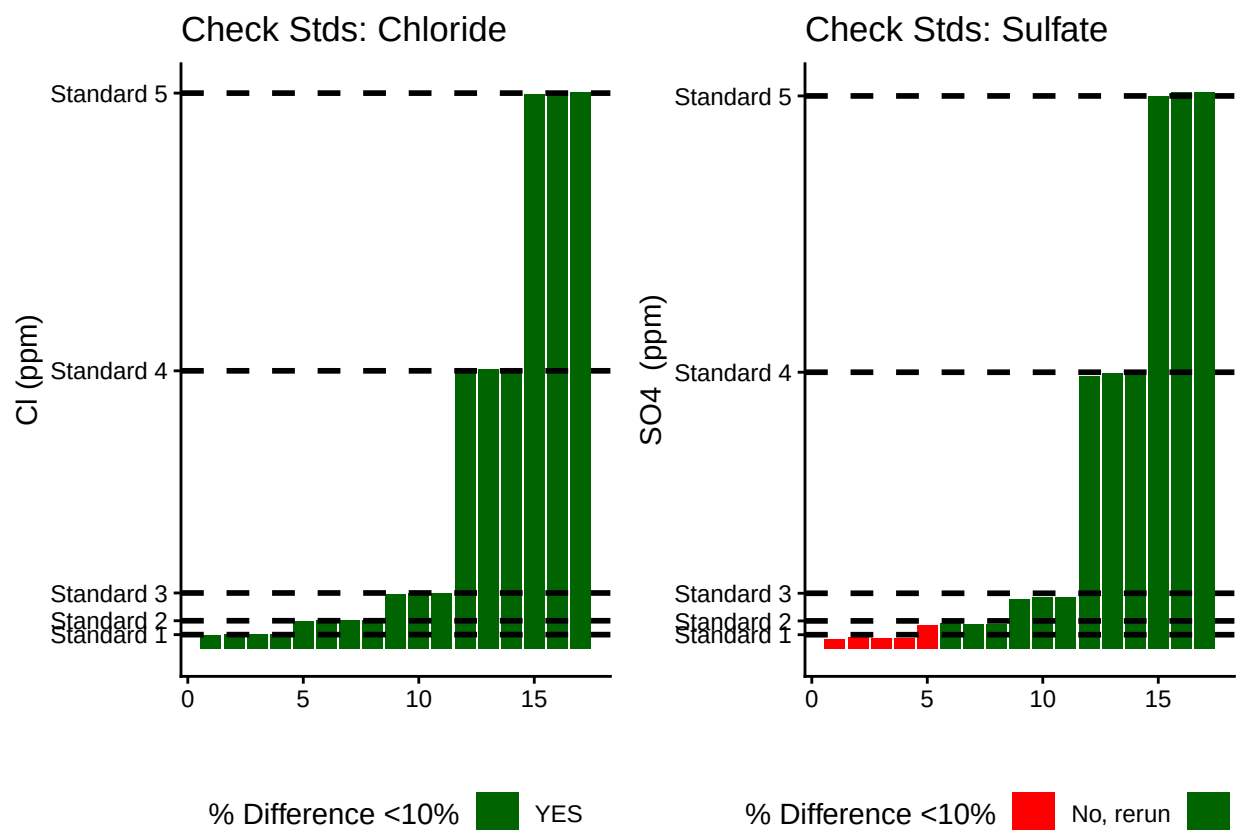
0.3 Assess Check Standards

```
## # A tibble: 5 x 5
##   sample_ID mean_Cl sd_Cl cv_Cl flag_Cl
##   <chr>      <dbl> <dbl> <dbl> <chr>
## 1 Standard 1    5.06 0.0349 0.00691 Chloride Check Standard RSD within Range - ~
## 2 Standard 2   10.1 0.0512 0.00508 Chloride Check Standard RSD within Range - ~
## 3 Standard 3   19.8 0.0847 0.00427 Chloride Check Standard RSD within Range - ~
## 4 Standard 4  100. 0.372 0.00371 Chloride Check Standard RSD within Range - ~
## 5 Standard 5  200. 0.587 0.00293 Chloride Check Standard RSD within Range - ~
```

```
## # A tibble: 5 x 5
##   sample_ID mean_S04 sd_S04 cv_S04 flag_S04
##   <chr>      <dbl> <dbl> <dbl> <chr>
## 1 Standard 1    0.392 0.0306 0.0780 Sulfate Check Standard RSD within Range - ~
## 2 Standard 2    0.891 0.0339 0.0380 Sulfate Check Standard RSD within Range - ~
## 3 Standard 3    1.84 0.0368 0.0200 Sulfate Check Standard RSD within Range - ~
## 4 Standard 4    9.93 0.0538 0.00542 Sulfate Check Standard RSD within Range - ~
## 5 Standard 5   20.1 0.0954 0.00475 Sulfate Check Standard RSD within Range - ~
```

```
## [1] ">80% of Chloride Check Standards have RSD within range - PROCEED"
```

```
## [1] ">80% of Sulfate Check Standards have RSD within range - PROCEED"
```



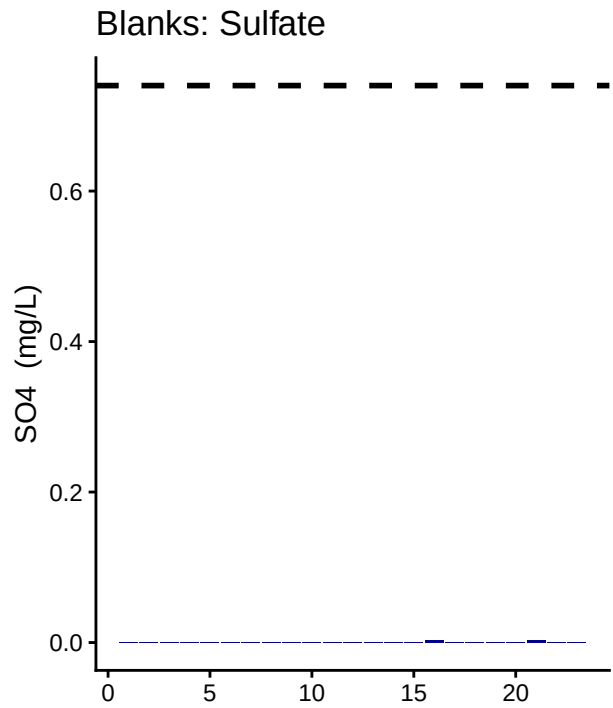
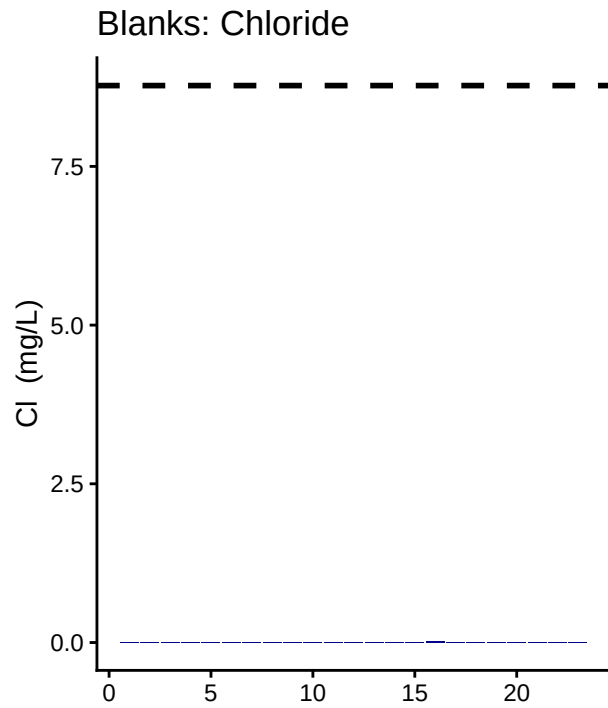
```
## [1] ">80% of Chloride Check Standards are within range of expected concentration - PROCEED"
```

```
## [1] "<80% of Sulfate Check Standards are within range of expected concentration - REASSESS"
```

0.4 Assess Blanks

```
## [1] ">80% of Chloride Blank concentrations are lower 25% quartile of samples"
```

```
## [1] ">80% of Sulfate Blank concentrations are lower 25% quartile of samples"
```



Blank Conc <25% Quartile Samples ■ YE

Blank Conc <25% Quartile Samples ■ YE

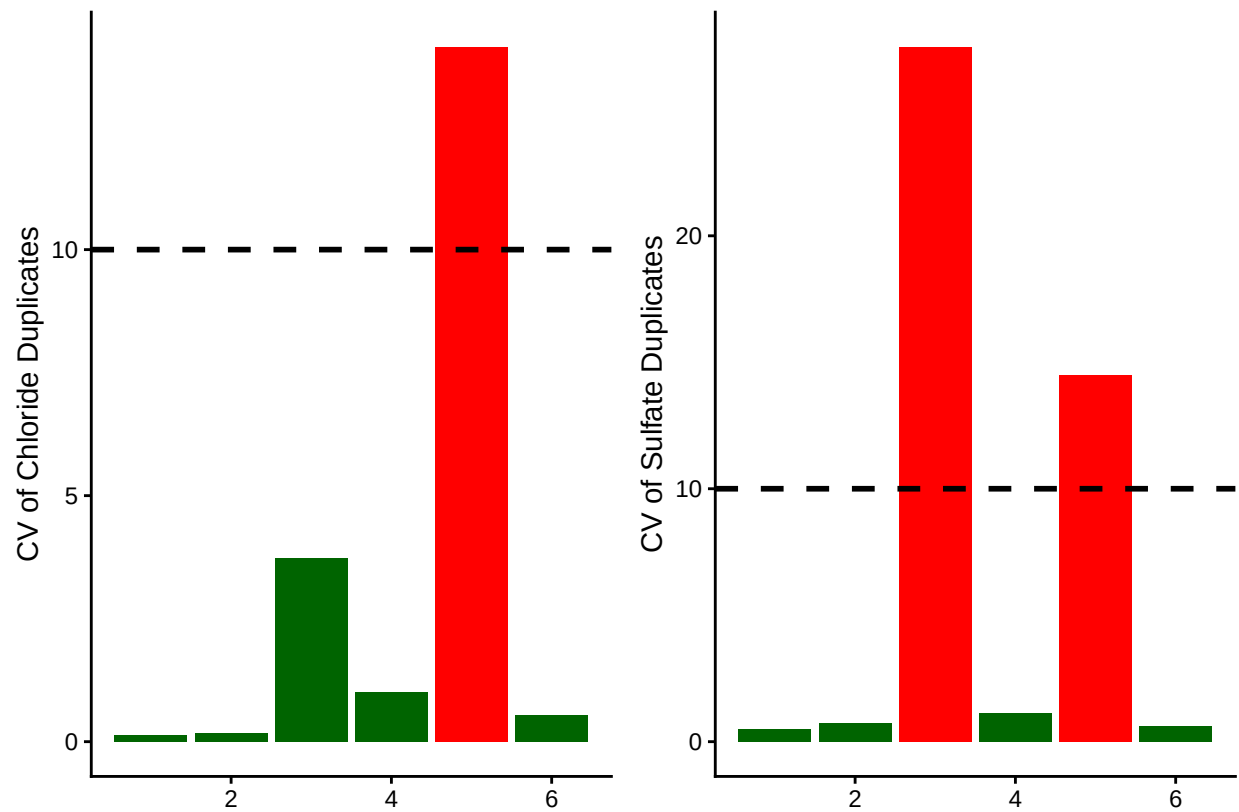
```
## Chloride blanks mean ppm:
```

```
## [1] 0.002308696
```

```
## Sulfate blanks mean ppm:
```

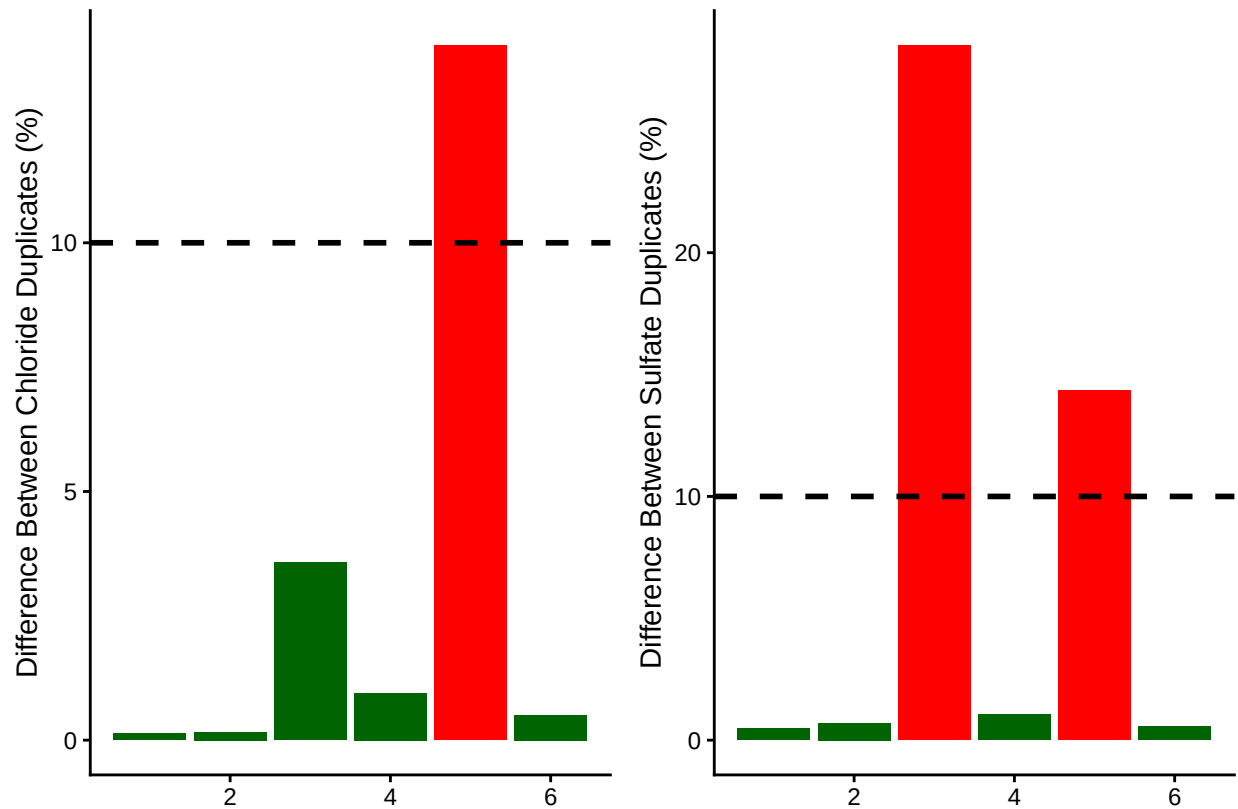
```
## [1] 0.0002913043
```

0.5 Assess Duplicates



```
## [1] ">80% of Chloride Duplicates have a CV <10% - PROCEED"
```

```
## [1] "<80% of Sulfate Duplicates have a CV <10% - REASSESS"
```



```
## [1] ">80% of Chloride Duplicates have a percent difference <10% - PROCEED"
```

```
## [1] "<80% of Sulfate Duplicates have a CV <10% - REASSESS"
```

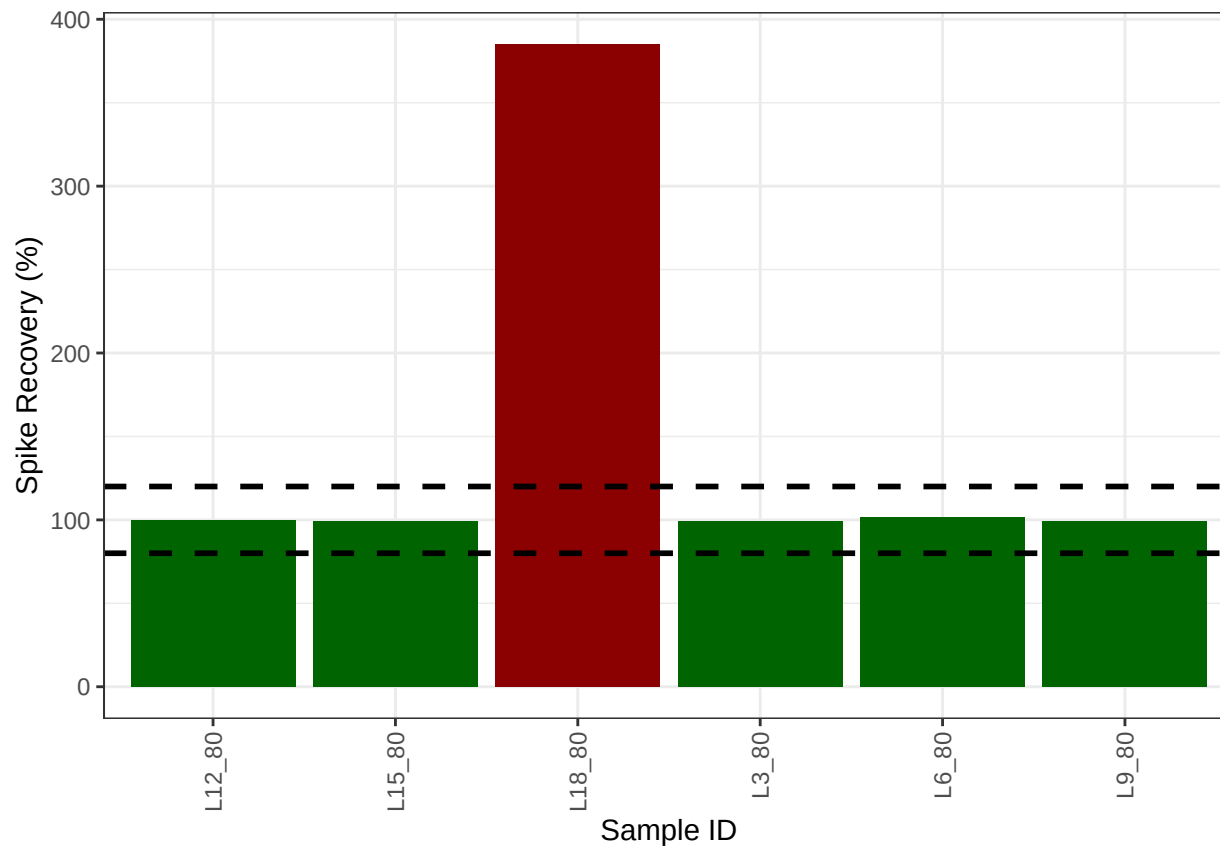
0.6 Calculate mmol/L concentrations & salinity, add dilutions

```
# Convert ppm to mmol/L
all_dat$S04_Conc_mM <- (all_dat$S04_ppm / s_mw)
all_dat$Cl_Conc_mM <- (all_dat$Cl_ppm / cl_mw)

# Calculate Salinity
# calculated using the Knudsen equation
# Salinity = 0.03 + 1.8050 * Chlorinity
# Ref: A Practical Handbook of Seawater Analysis by Strickland & Parsons (P. 11)
# =((1.807*Cl_ppm)+0.026)/1000
all_dat$salinity <- ((1.8070 * all_dat$Cl_ppm) + 0.026) / 1000

#Need to determine dilution factors for your samples
#for Steph / COMPASS this depends on the site so...
all_dat$Dilution <- Dilution
# head(all_dat)
```

0.7 Assess Analytical Spikes



```
## [1] ">80% of S04 spikes have a recovery between the high and low cutoff - PROCEED"
```

0.8 Check if samples within the range of the standard curve

```
## Sample Flagging
```

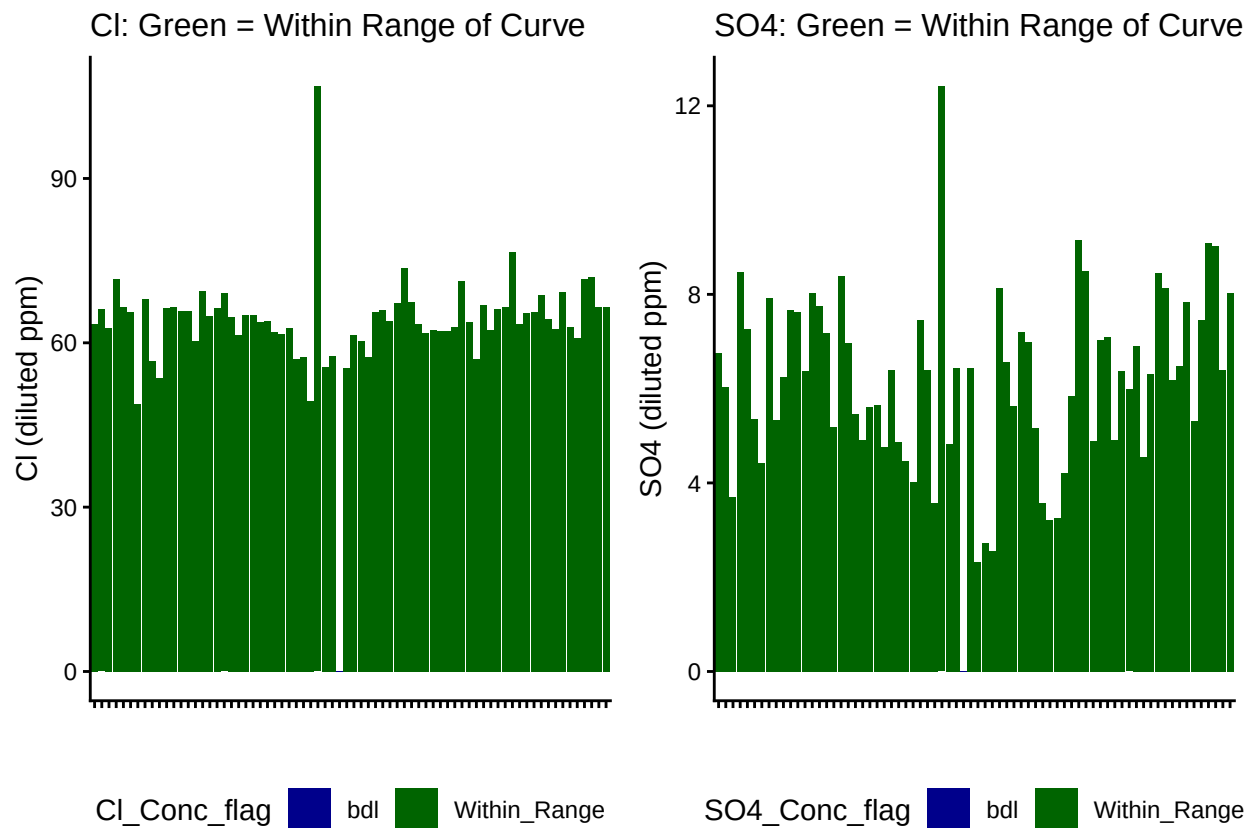


Table 1: SO4 samples

SO4_Conc_flag	Percent_samples
Within_Range	98.611111
bdl	1.388889

Table 2: Cl samples

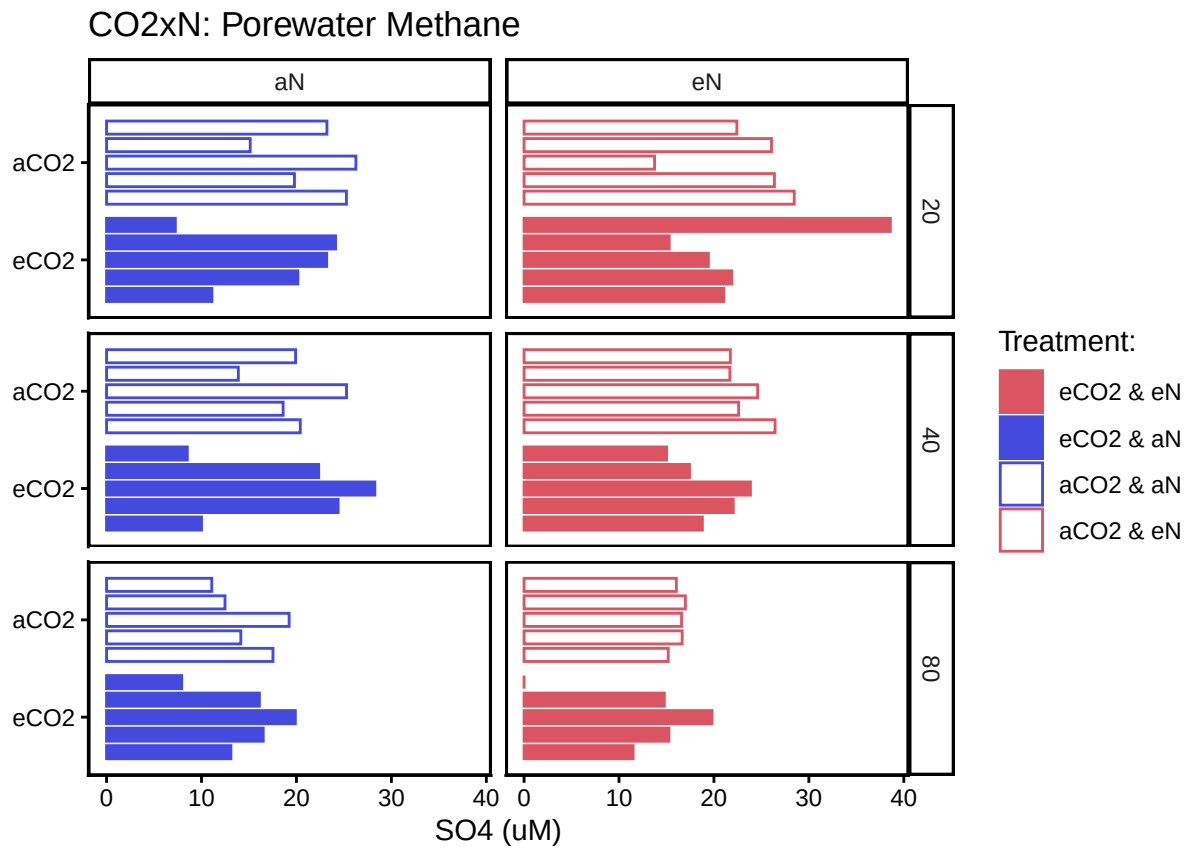
Cl_Conc_flag	Percent_samples
Within_Range	98.611111
bdl	1.388889

0.9 Check to see if samples run match metadata & merge info

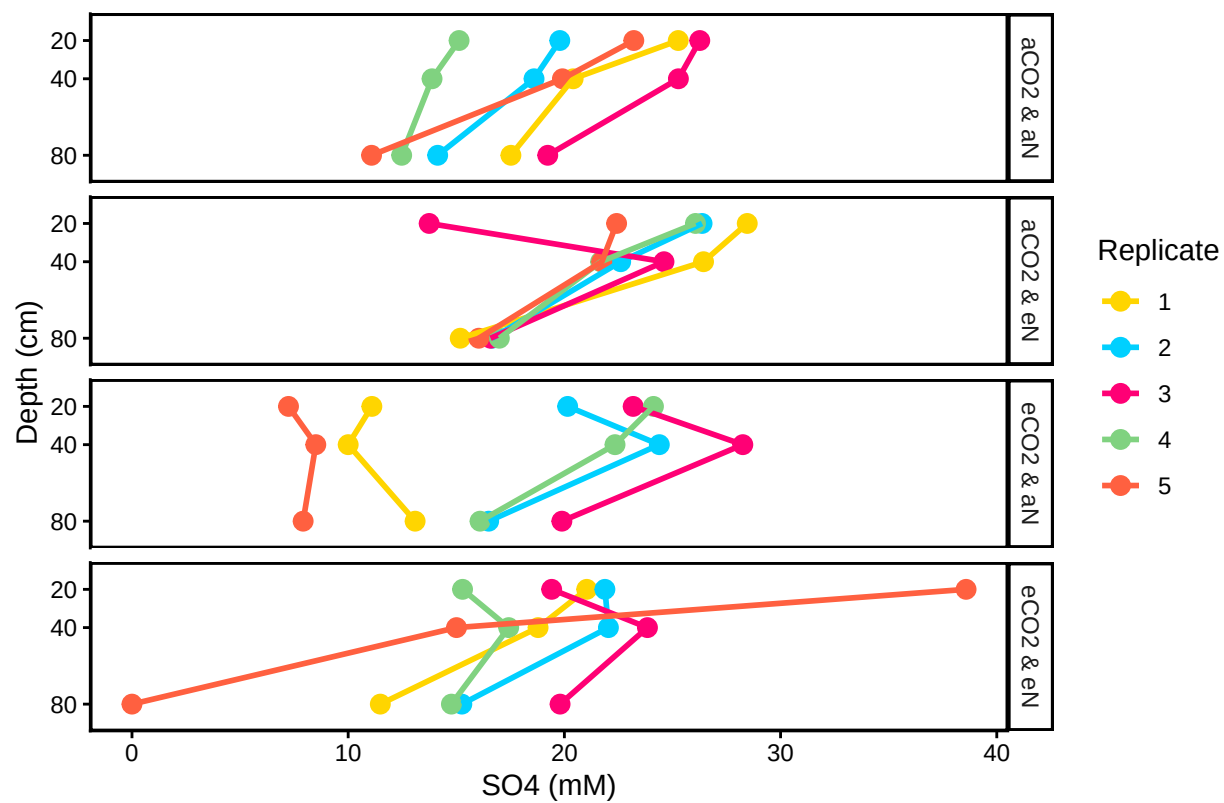
```
## All sample IDs are present in metadata.
```

0.10 Organize Data

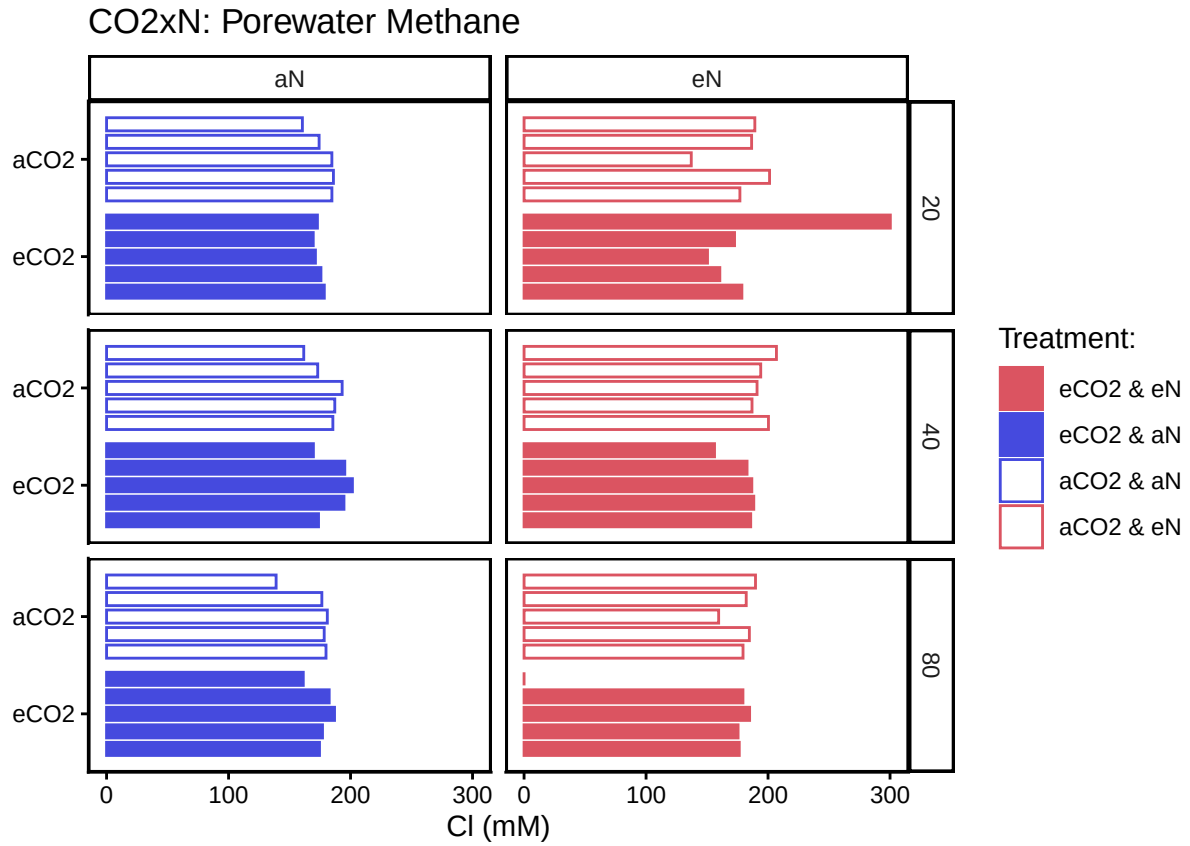
0.11 Visualize Data SO4

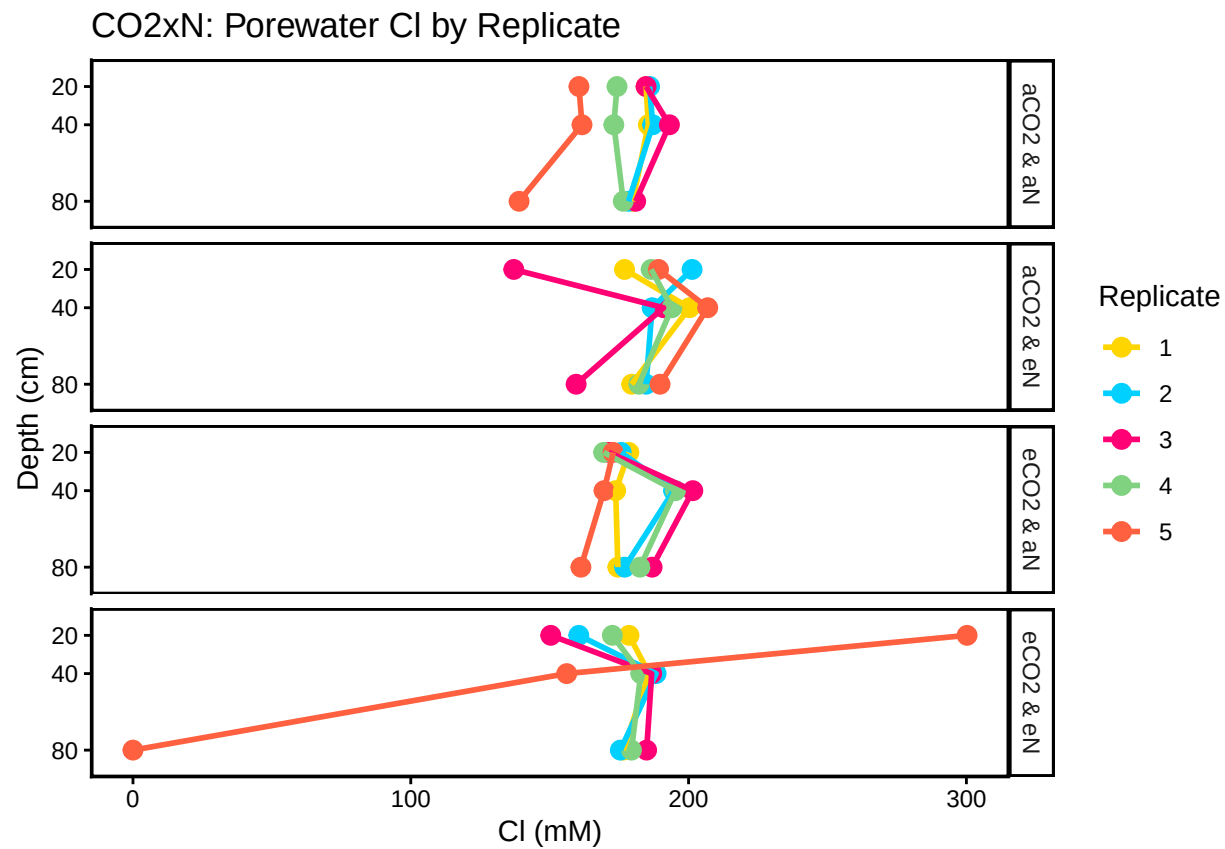


CO2xN: Porewater SO4 by Replicate



0.12 Visualize Data CI





0.13 Export Processed Data

#end